Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| Title of Proposed Technology | Advanced Coatings for Beyond-Extreme Ultraviolet Lithography (BEUVL) | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | **Electrical・Electronics** | | Electrical・Electronics | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Semiconductor Equipment | Core Components and Manufacturing Equipment for Semiconductor Tools | Optical Application Devices | Optical Components |
| Concept of Proposed Technology | The proposed technology builds on existing BEUV mirror coatings operating at approximately 6.X nm and focuses on their further advancement for BEUV lithography. The core objectives are the improvement of optical performance as well as thermal and temporal stability under application-relevant conditions. Advanced metrology is used as a key tool to assess current performance limits and support the systematic development of stable and lithography-ready BEUV mirror technology.<br><br>Improvement approaches for reflectivity and stability are addressed in an application-oriented and technology-agnostic manner, including:<br>• Exploration of advanced coating material concepts to enhance optical efficiency and long-term stability.<br>• Interface and barrier engineering approaches, including interface smoothing strategies, to | | | |

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| | control interface quality and suppress interdiffusion processes in multilayer interference mirrors.<br>• Application of modern thin-film coating technologies, such as plasma-based, ion-assisted, and atomic-scale deposition approaches, to enable improved layer quality and reproducibility. |
| **Rationale for Joint R&D Project** | BEUV mirror technology operating at approximately 6.X nm is currently available only at a limited number of research institutions, with Fraunhofer holding established expertise and existing mirror solutions [1-3]. In Korea, such BEUV mirror technology is not yet available, while strong capabilities exist in semiconductor manufacturing, lithography system development, and industrial application of EUV technologies. Joint R&D is therefore required to transfer and adapt advanced BEUV mirror technology to lithography-relevant environments, to align mirror performance and stability with system-level requirements, and to enable future industrial adoption within the Korean semiconductor ecosystem. |
| **Key Words** | 1. Next-generation lithography<br>2. BEUV mirror technology<br>3. 6.X nm wavelength<br>4. Semiconductor manufacturing<br>5. Advanced metrology |

## 3. Research and Development Objectives & Plan

| **Technology Readiness Level** (TRL)[2] | **Start of the Project** | Level (4) | **End of the Project** | Level (8) |
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| **R&D Objectives** | The R&D project aims to advance existing BEUV mirror technology at 6.X nm toward lithography-relevant performance. Key objectives include the improvement of optical efficiency, enhancement of thermal and temporal stability under application-relevant conditions, and the qualification of BEUV mirror coatings for future BEUV lithography. Supporting objectives include the establishment of reliable metrology and evaluation procedures to assess technology maturity and stability. | | | |
| **R&D Target Specifications** | The R&D project targets the improvement of BEUV mirror performance at a wavelength of approximately 6.X nm for lithography-relevant applications. The target specifications include:<br>• Peak reflectivity: ≥ 65% at 6.X nm (near-normal incidence).<br>• Spectral bandwidth: increased bandwidth of $\Delta\lambda/\lambda \geq 1.0\%$ (FWHM) to overcome the current narrow-band limitation of BEUV mirrors.<br>• Thermal stability: reflectivity degradation ≤ 1% (absolute) and peak-wavelength shift ≤ 0.01 nm after thermal loading under application-relevant conditions.<br>• Temporal stability: reflectivity degradation ≤ 1% (absolute) over extended time periods under defined storage and operation conditions. | | | |
| **R&D Plan** | Phase I – Pre-study (0–6 months):<br>• Definition of lithography-relevant performance and stability requirements for BEUV mirrors at 6.X nm.<br>• Baseline characterization of existing BEUV mirror coatings using advanced optical and structural metrology.<br>• Identification of key performance-limiting factors affecting optical efficiency, spectral bandwidth, and stability.<br>• Development of an R&D roadmap and specification set for targeted performance improvements.<br>Milestones Phase I:<br>• Reference performance dataset for existing BEUV mirrors.<br>• Defined target specifications and evaluation criteria for Phase II.<br>• Feasibility assessment for performance and stability enhancement.<br>Phase II – Main R&D (7–36 months):<br>• Systematic improvement of BEUV mirror performance with focus on optical efficiency, bandwidth broadening, and thermal and temporal stability.<br>• Comparative evaluation of mirror concepts and design approaches under application-relevant conditions.<br>• Implementation of reliability and stability testing under thermal load and extended operation scenarios.<br>• Continuous metrology-based feedback to guide optimization and assess technology maturity.<br>• Final qualification of BEUV mirror technology against lithography-relevant target specifications.<br>Milestones Phase II:<br>• Demonstration of BEUV mirrors meeting target reflectivity and bandwidth specifications.<br>• Verified thermal and temporal stability under defined conditions.<br>• Technology readiness assessment for future BEUV lithography integration. | | | |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | The research institute (Fraunhofer) will be responsible for the core technical development and evaluation of BEUV mirror technology at 6.X nm. The main roles and responsibilities include:<br>• Development and optimization of BEUV mirror coatings toward improved optical performance and stability for lithography-relevant applications.<br>• Execution of advanced optical, structural, and stability metrology to assess mirror performance, bandwidth, and reliability.<br>• Design and implementation of thermal and temporal stability testing under application-relevant conditions.<br>• Comparative evaluation of mirror concepts and systematic performance assessment based on defined target specifications.<br>• Provision of research infrastructure, deposition and characterization facilities, and scientific expertise required for BEUV mirror development.<br>• Contribution to joint project coordination, reporting, and technical exchange within the international R&D consortium.<br>• The coating technology will be prepared for transfer to Korea for integration into BEUV optical platforms (TBD). |
| **Expected R&R of Korean SME Partner** | The Korean SME partner is expected to contribute application-driven expertise and infrastructure relevant to BEUV lithography and semiconductor manufacturing. Key roles and responsibilities include:<br>• Provision of BEUV irradiation capabilities at relevant wavelengths to enable radiation stability testing of BEUV mirror samples.<br>• Execution of irradiation-based degradation and reliability tests under application-relevant exposure conditions.<br>• Contribution to the definition of lithography-relevant requirements, testing protocols, and validation criteria.<br>• Support of system-level evaluation, integration aspects, and industrial feasibility assessment.<br>• Provision of technical resources, test infrastructure, and engineering support required for joint R&D activities.<br>• Active participation in technical exchange, data evaluation, and joint project coordination within the consortium. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | The joint R&D project is expected to deliver BEUV coating technology at 6.X nm with improved optical performance, broadened spectral bandwidth, and enhanced thermal and temporal stability suitable for lithography-relevant applications. These results will contribute to the qualification of BEUV mirrors as a key enabling component for future BEUV lithography.<br><br>From a commercialization perspective, the project will establish a technological basis for the transfer of high-performance mirror coating technology for beyond-EUV lithography into industrial semiconductor environments. The developed coating technology will be prepared for transfer to Korea, enabling future integration into lithography tools, inspection systems, and related BEUV optical platforms. The outcomes support the development of next-generation lithography technologies beyond current EUV systems.<br><br>At the industrial and economic level, the project strengthens international cooperation between Fraunhofer and the Korean semiconductor ecosystem, supports Korea's strategic positioning in next-generation lithography, and contributes to the long-term competitiveness of the semiconductor industry through access to critical BEUV optical technologies. |

## 6. References

(1) Naujok P., Yulin, S., Müller R., Kaiser N., Tünnermann A.: Interface characterization in B-based multilayer mirrors for next generation lithography: Thin Solid Films, 2016, Vol.612, pp.414-418.

(2) Yulin S., Trost M, Schwinde St., Schröder S.: Coatings with barrier layers for extreme short wavelengths: VIP, 2021, Vol. 33, 5, pp. 24 - 29.

(3) Naujok P., Murray K., Yulin S., Kaiser N.: Mirror for reflecting EUV radiation with tension compensation and method for producing the same. EP 3239745 B1 / DE10 2016 118 940.

## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |